

RAM AND STORAGE THEORY

1.Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Ans.

Feature	DDR3	DDR4	DDR5
Speed	Around 800–2133 MT/s	Around 1600–3200 MT/s	Around 3200–8400+ MT/s
Voltage	1.5 V	1.2V	1.1 V
Performance	Normal	Better than DDR3	Faster than DDR4
Pin Count (DIMM / SODIMM)	240-pin (Desktop) / 204-pin (Laptop)	288-pin (Desktop) / 260-pin (Laptop)	288-pin (Desktop) / 262-pin (Laptop)
Power Consumption	Comparatively higher	Lower than DDR3	Lower voltage and better power efficiency
Typical Use	Old desktop computers and laptops	Many desktop PCs and laptops	Newer and high-performance computers

2. Locate the RAM slot in your computer, take a clear photo of the RAM module, reinstall it properly, and check whether the computer starts successfully.

Ans.

Steps:

Step 1 : First, I shut down the computer and disconnected the power supply.

Step 2 : I opened the computer and located the RAM slot.

Step 3 : I carefully removed the RAM module from the slot.

Step 4 : I took a clear photo of the RAM module.

Step 5 : I inserted the RAM module back into the same slot properly.

Step 6 : I closed the computer and connected the power supply.

Step 7 : I turned on the computer and checked that it booted successfully.

Step 8 : Finally, I verified that the RAM was detected by the system.

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2. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Ans.

Feature	HDD	SSD	NVMe
Speed	slower	Faster than HDD	Very fast
technology	Uses spinning disks	Uses flash memory	Uses flash memory with PCIe
Use case	File storage and backup	Daily computer use	Gaming and heavy applications
Daily app/use		Using Chrome and other daily apps	Playing games with faster loading

4. Remove the storage drive, identify and photograph its connector type (SATA, M.2, or NVMe), reinstall it, and check that the computer detects the drive.

Ans.

Steps:

Step 1 : I shut down the computer and disconnected the power.

Step 2 : I opened the computer and located the storage drive.

Step 3 : I carefully removed the HDD/SSD from its slot.

Step 4 : I checked the connector type and took a clear photo.

Step 5 : I reinstalled the storage drive properly.

Step 6 : I closed the computer and connected the power.

Step 7 : I started the computer and checked that the storage drive was detected successfully.